

SHETH L.U.J. AND SIR M.V. COLLEGE

Aim: Web Security with SSL/TLS

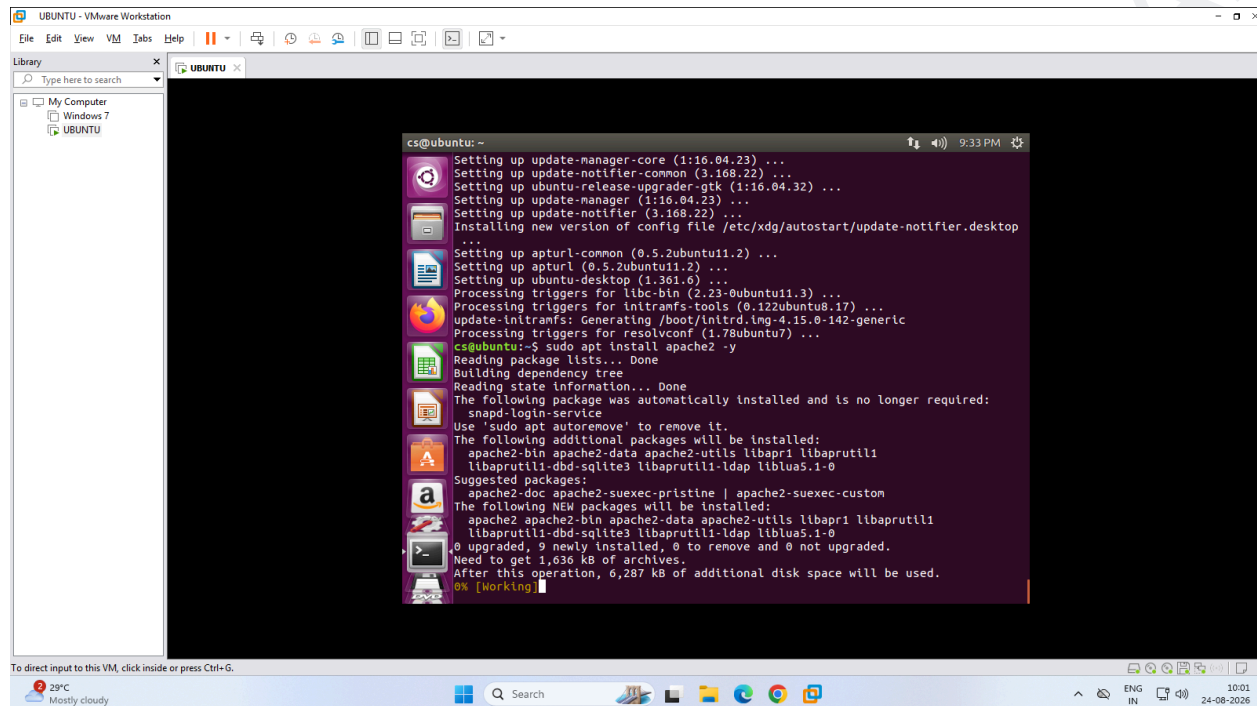
Step 1: Update the System Open Terminal and run:

```
sudo apt update
```

```
sudo apt upgrade -y
```

Step 2: Install Apache Web Server

```
sudo apt install apache2 -y
```



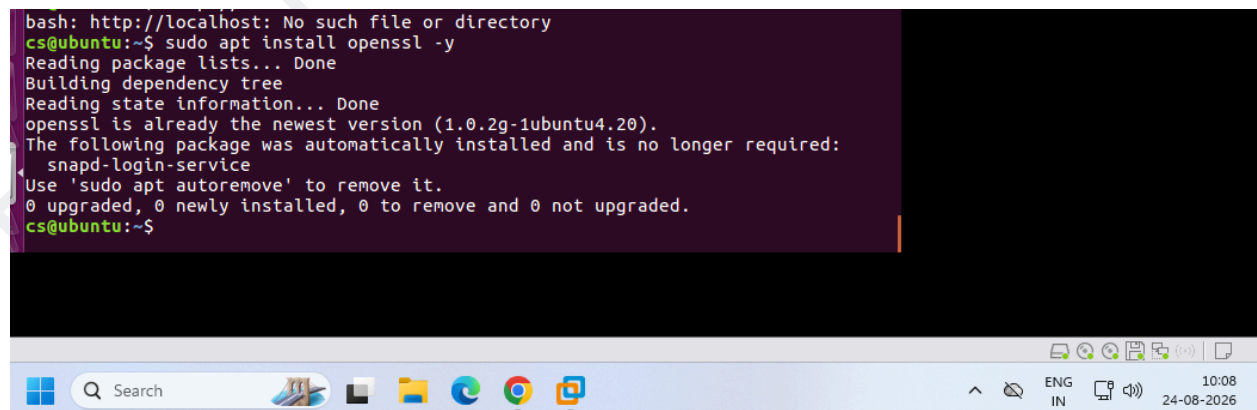
The screenshot shows a terminal window titled 'cs@ubuntu: ~' with the following output:

```
Setting up update-manager-core (1:16.04.23) ...
Setting up update-notifier-common (3:16.8.22) ...
Setting up ubuntu-release-upgrader-gtk (1:16.04.32) ...
Setting up update-manager (1:16.04.23) ...
Setting up update-notifier (3:16.8.22) ...
Installing new version of config file /etc/xdg/autostart/update-notifier.desktop
...
Setting up apturl-common (0.5.2ubuntu11.2) ...
Setting up apturl (0.5.2ubuntu11.2) ...
Setting up ubuntu-desktop (1.361.6) ...
Processing triggers for libc-bin (2.23-0ubuntu11.3) ...
Processing triggers for initramfs-tools (0.122ubuntu8.17) ...
update-initramfs: Generating /boot/initrd.img-4.15.0-142-generic
Processing triggers for resolvconf (1.78ubuntu7) ...
cs@ubuntu:~$ sudo apt install apache2 -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following package was automatically installed and is no longer required:
  snapd-login-service
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils libapr1 libaprutil1
  libaprutil1-dbd-sqlite3 libaprutil1-ldap liblua5.1-0
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapr1 libaprutil1
  libaprutil1-dbd-sqlite3 libaprutil1-ldap liblua5.1-0
0 upgraded, 9 newly installed, 0 to remove and 0 not upgraded.
Need to get 1,636 kB of archives.
After this operation, 6,287 kB of additional disk space will be used.
0% [Working]
```

Step 3: Install OpenSSL OpenSSL is used to create and manage SSL/TLS certificates.

```
sudo apt install openssl -y
```

Check the version: openssl version



The screenshot shows a terminal window with the following output:

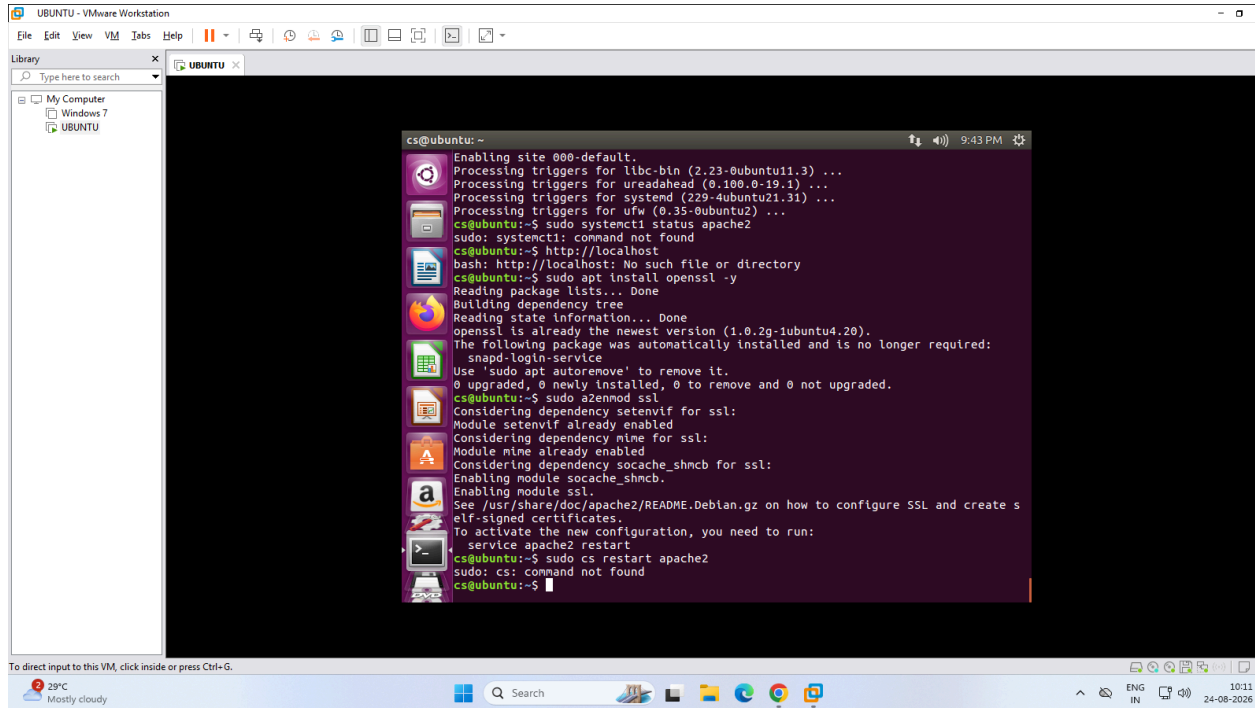
```
bash: http://localhost: No such file or directory
cs@ubuntu:~$ sudo apt install openssl -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
openssl is already the newest version (1.0.2g-1ubuntu4.20).
The following package was automatically installed and is no longer required:
  snapd-login-service
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
cs@ubuntu:~$
```

Step 4: Enable SSL Module in Apache

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Restart Apache:

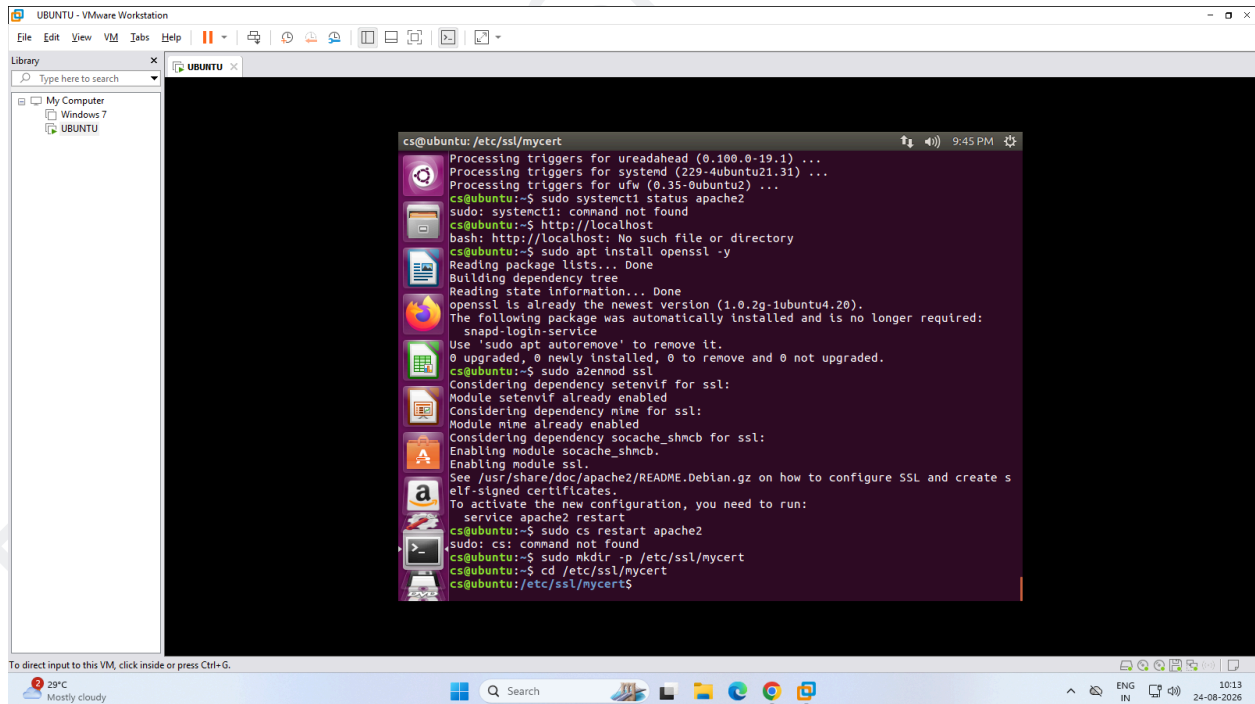
`sudo systemctl restart apache2`



```
cs@ubuntu:~$ sudo systemctl status apache2
sudo: systemctl: command not found
cs@ubuntu:~$ http://localhost
bash: http://localhost: No such file or directory
cs@ubuntu:~$ sudo apt install openssl -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
openssl is already the newest version (1.0.2g-1ubuntu4.20).
The following package was automatically installed and is no longer required:
  snapd-login-service
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
cs@ubuntu:~$ sudo a2enmod ssl
Considering dependency setenvif for ssl:
Module setenvif already enabled
Considering dependency mime for ssl:
Module mime already enabled
Considering dependency socache_shmcb for ssl:
Enabling module socache_shmcb.
Enabling module ssl.
See /usr/share/doc/apache2/README.Debian.gz on how to configure SSL and create s
elf-signed certificates.
To activate the new configuration, you need to run:
  service apache2 restart
cs@ubuntu:~$ sudo cs restart apache2
sudo: cs: command not found
cs@ubuntu:~$
```

Step 5: Create a Directory for Certificates

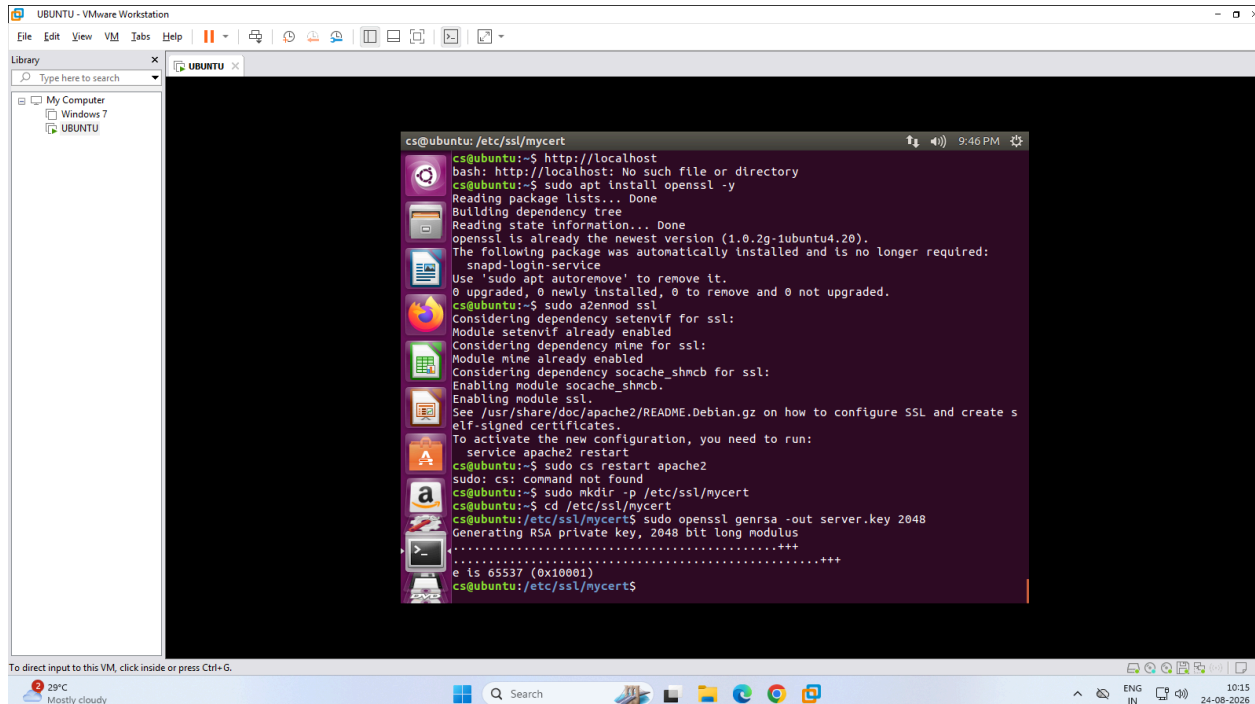
`sudo mkdir -p /etc/ssl/mycert`



```
cs@ubuntu:/etc/ssl/mycert$
Processing triggers for ureadahead (0.100.0-19.1) ...
Processing triggers for systemd (229-4ubuntu21.31) ...
cs@ubuntu:~$ sudo systemctl status apache2
sudo: systemctl: command not found
cs@ubuntu:~$ http://localhost
bash: http://localhost: No such file or directory
cs@ubuntu:~$ sudo apt install openssl -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
openssl is already the newest version (1.0.2g-1ubuntu4.20).
The following package was automatically installed and is no longer required:
  snapd-login-service
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
cs@ubuntu:~$ sudo a2enmod ssl
Considering dependency setenvif for ssl:
Module setenvif already enabled
Considering dependency mime for ssl:
Module mime already enabled
Considering dependency socache_shmcb for ssl:
Enabling module socache_shmcb.
Enabling module ssl.
See /usr/share/doc/apache2/README.Debian.gz on how to configure SSL and create s
elf-signed certificates.
To activate the new configuration, you need to run:
  service apache2 restart
cs@ubuntu:~$ sudo cs restart apache2
sudo: cs: command not found
cs@ubuntu:~$ sudo mkdir -p /etc/ssl/mycert
cs@ubuntu:~$ cd /etc/ssl/mycert
cs@ubuntu:/etc/ssl/mycert$
```

Step 6: Generate a Private Key

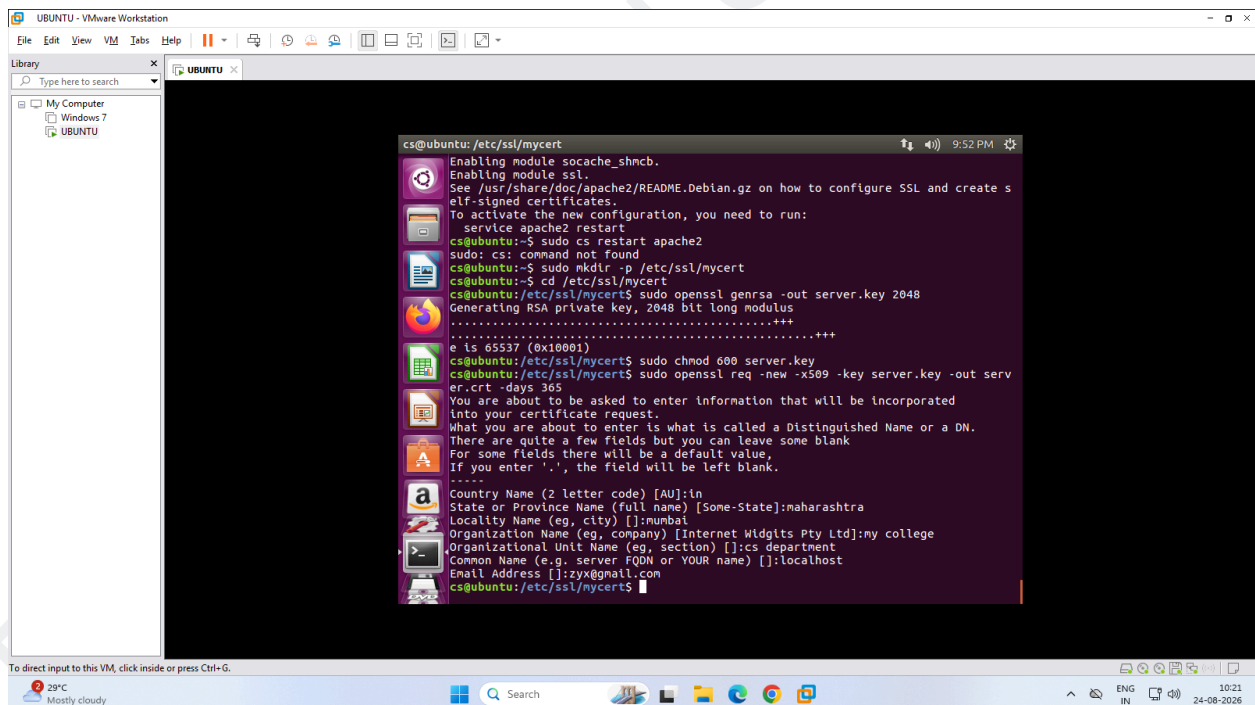
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```
cs@ubuntu:/etc/ssl/mycert
cs@ubuntu:~$ http://localhost
bash: http://localhost: No such file or directory
cs@ubuntu:~$ sudo apt install openssl -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
openssl is already the newest version (1.0.2g-1ubuntu4.20).
The following package was automatically installed and is no longer required:
  snapd-login-service
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
cs@ubuntu:~$ sudo a2enmod ssl
Considering dependency setenvif for ssl:
Module setenvif already enabled
Considering dependency mime for ssl:
Module mime already enabled
Considering dependency socache_shmcb for ssl:
Enabling module socache_shmcb.
Enabling module ssl.
See /usr/share/doc/apache2/README.Debian.gz on how to configure SSL and create s
elf-signed certificates.
To activate the new configuration, you need to run:
  service apache2 restart
cs@ubuntu:~$ sudo cs restart apache2
sudo: cs: command not found
cs@ubuntu:~$ sudo mkdir -p /etc/ssl/mycert
cs@ubuntu:~$ cd /etc/ssl/mycert
cs@ubuntu:/etc/ssl/mycert$ sudo openssl genrsa -out server.key 2048
Generating RSA private key, 2048 bit long modulus
.....+++++
e is 65537 (0x10001)
cs@ubuntu:/etc/ssl/mycert$
```

Step 7: Generate a Self-Signed SSL Certificate Run:

`sudo openssl req -new -x509 -key server.key -out server.crt -days 365`



```
cs@ubuntu:/etc/ssl/mycert
Enabling module socache_shmcb.
Enabling module ssl.
See /usr/share/doc/apache2/README.Debian.gz on how to configure SSL and create s
elf-signed certificates.
To activate the new configuration, you need to run:
  service apache2 restart
cs@ubuntu:~$ sudo cs restart apache2
sudo: cs: command not found
cs@ubuntu:~$ sudo mkdir -p /etc/ssl/mycert
cs@ubuntu:~$ cd /etc/ssl/mycert
cs@ubuntu:/etc/ssl/mycert$ sudo openssl genrsa -out server.key 2048
Generating RSA private key, 2048 bit long modulus
.....+++++
e is 65537 (0x10001)
cs@ubuntu:/etc/ssl/mycert$ sudo chmod 600 server.key
cs@ubuntu:/etc/ssl/mycert$ sudo openssl req -new -x509 -key server.key -out serv
er.crt -days 365
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:in
State or Province Name (full name) [Some-State]:maharashtra
Locality Name (eg, city) []:mumbai
Organization Name (eg, company) [Internet Hldgts Pty Ltd]:my college
Organizational Unit Name (eg, section) []:cs department
Common Name (e.g. server FQDN or YOUR name) []:localhost
Email Address []:zyx@gmail.com
cs@ubuntu:/etc/ssl/mycert$
```

Step 8: Configure Apache for HTTPS

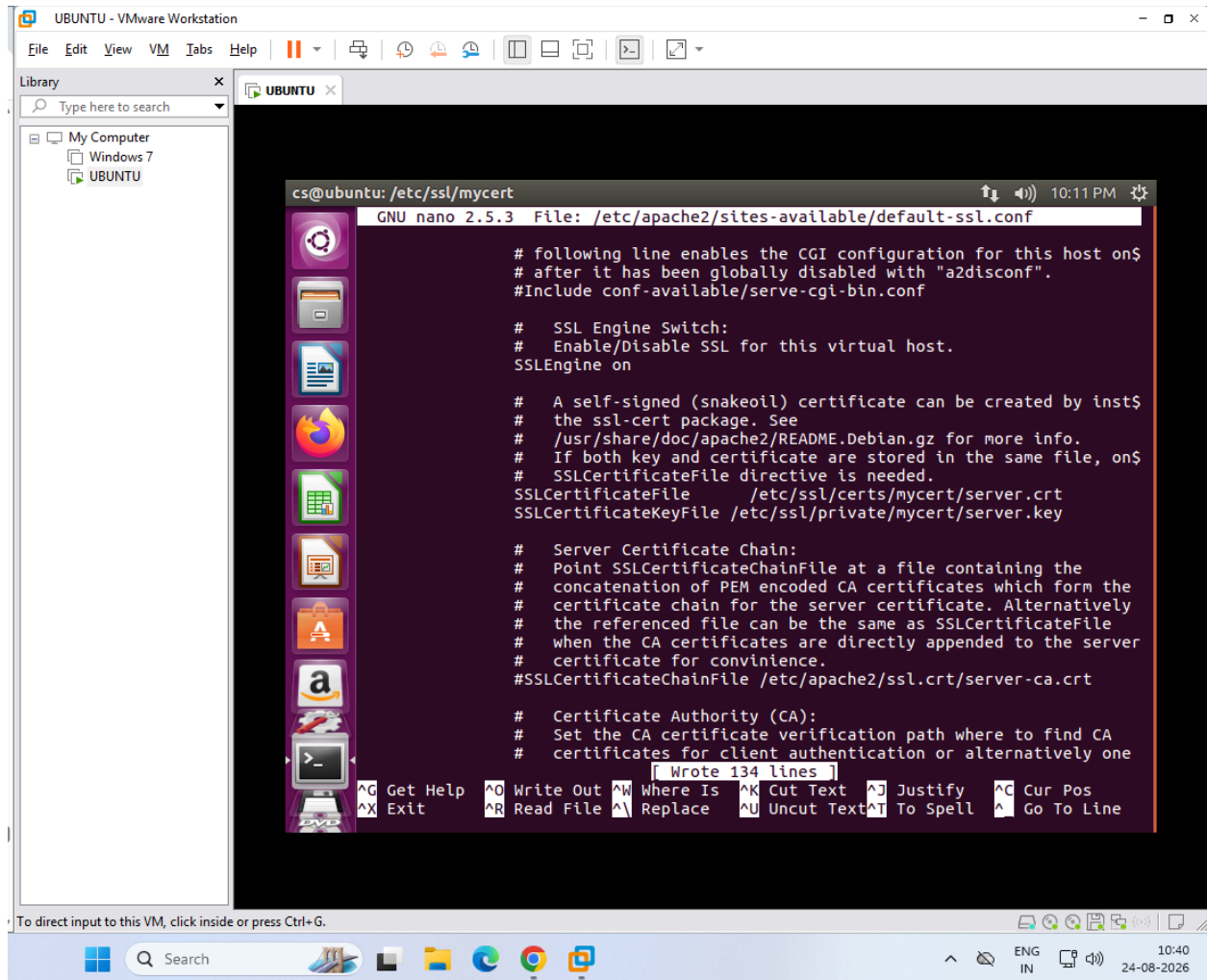
Enable the default SSL site:

`sudo a2ensite default-ssl`

Edit the SSL configuration:

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sudo nano /etc/apache2/sites-available/default-ssl.conf



```
cs@ubuntu: /etc/ssl/mycert$ nano /etc/apache2/sites-available/default-ssl.conf
GNU nano 2.5.3 File: /etc/apache2/sites-available/default-ssl.conf

# following line enables the CGI configuration for this host on$
# after it has been globally disabled with "a2disconf".
#Include conf-available/serve-cgi-bin.conf

# SSL Engine Switch:
# Enable/Disable SSL for this virtual host.
SSLEngine on

# A self-signed (snakeoil) certificate can be created by inst$
# the ssl-cert package. See
# /usr/share/doc/apache2/README.Debian.gz for more info.
# If both key and certificate are stored in the same file, on$
# SSLCertificateFile directive is needed.
SSLCertificateFile /etc/ssl/certs/mycert/server.crt
SSLCertificateKeyFile /etc/ssl/private/mycert/server.key

# Server Certificate Chain:
# Point SSLCertificateChainFile at a file containing the
# concatenation of PEM encoded CA certificates which form the
# certificate chain for the server certificate. Alternatively
# the referenced file can be the same as SSLCertificateFile
# when the CA certificates are directly appended to the server
# certificate for convinience.
#SSLCertificateChainFile /etc/apache2/ssl.crt/server-ca.crt

# Certificate Authority (CA):
# Set the CA certificate verification path where to find CA
# certificates for client authentication or alternatively one
# certificate for convinience.
#CAPath /etc/ssl/certs

[ Wrote 134 lines ]

^G Get Help ^O Write Out ^W Where Is ^K Cut Text ^J Justify ^C Cur Pos
^X Exit ^R Read File ^A Replace ^U Uncut Text ^T To Spell ^_ Go To Line
```

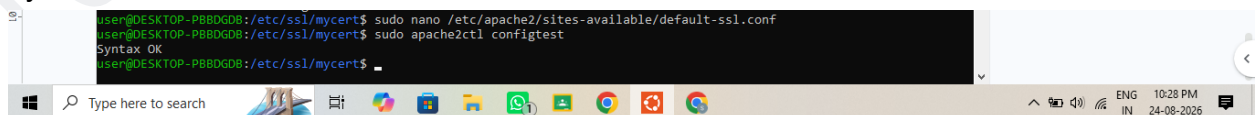
Step 9: Test Apache Configuration

Run:

sudo apache2ctl configtest

Expected result:

Syntax OK



```
user@DESKTOP-P8B0GDB:/etc/ssl/mycert$ sudo nano /etc/apache2/sites-available/default-ssl.conf
user@DESKTOP-P8B0GDB:/etc/ssl/mycert$ sudo apache2ctl configtest
Syntax OK
user@DESKTOP-P8B0GDB:/etc/ssl/mycert$
```

Step 10: Restart Apache

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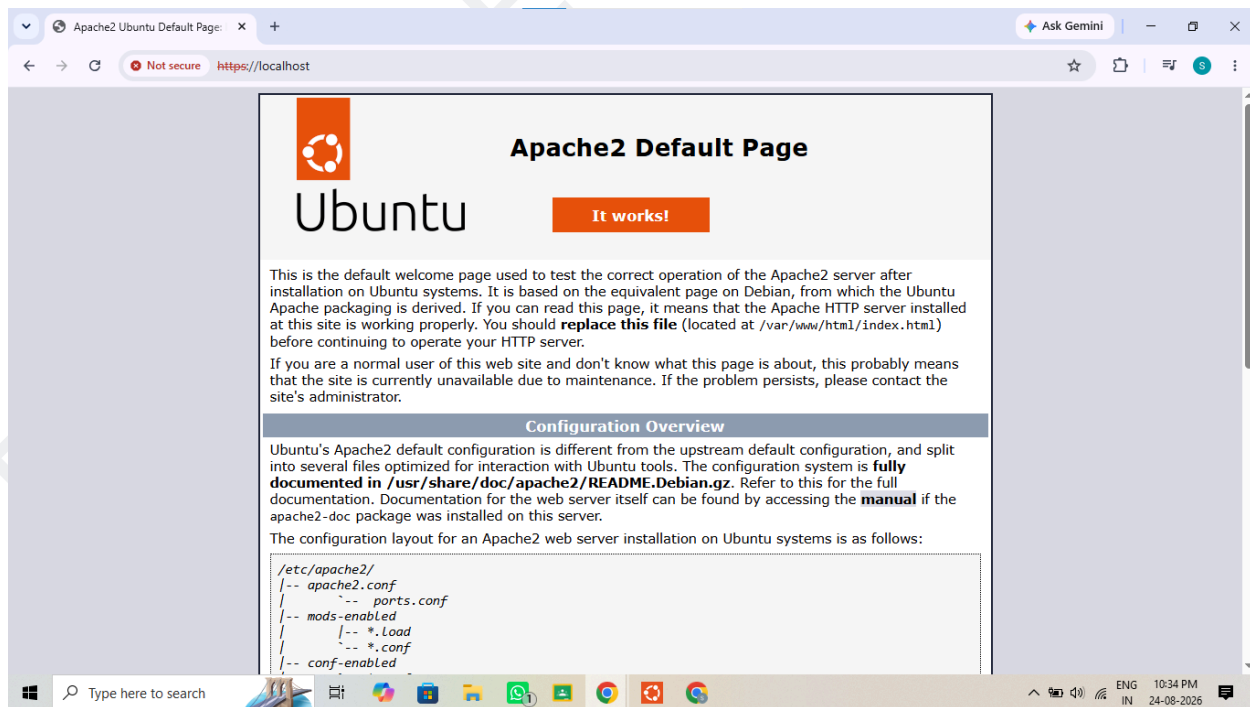
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```
user@DESKTOP-PB8DGD8: /etc/ssl/mycert$
H00526: Syntax error on line 31 of /etc/apache2/sites-enabled/default-ssl.conf:
SSLCertificateFile: file '/etc/ssl/certs/mycert/server.crt' does not exist or is empty
user@DESKTOP-PB8DGD8: /etc/ssl/mycert$ sudo sed -i 's|/etc/ssl/certs/mycert|/etc/ssl/mycert/g' /etc/apache2/sites-available/default-ssl.conf
user@DESKTOP-PB8DGD8: /etc/ssl/mycert$ grep -n "SSLCertificate" /etc/apache2/sites-available/default-ssl.conf
4: # SSLCertificateFile directive is needed.
1: SSLCertificateFile /etc/ssl/mycert/server.crt
2: SSLCertificateKeyFile /etc/ssl/private/mycert/server.key
3: # Point SSLCertificateChainFile at a file containing the
4: # the referenced file can be the same as SSLCertificateFile
1: #SSLCertificateChainFile /etc/apache2/ssl.crt/server-ca.crt
user@DESKTOP-PB8DGD8: /etc/ssl/mycert$ sudo apache2ctl configtest
H00526: Syntax error on line 31 of /etc/apache2/sites-enabled/default-ssl.conf:
SSLCertificateFile: file '/etc/ssl/mycert/server.crt' does not exist or is empty
user@DESKTOP-PB8DGD8: /etc/ssl/mycert$ sudo sed -i 's|SSLCertificateFile /etc/ssl/mycert/server.crt|; s|SSLCertificateKeyFile.*|SSLCertificateKeyFile
/etc/ssl/mycert/server.key|' /etc/apache2/sites-available/default-ssl.conf
sed: can't read /etc/apache2/sites-available/default-ssl.conf: No such file or directory
user@DESKTOP-PB8DGD8: /etc/ssl/mycert$ ls -l /etc/apache2/sites-available/
total 12
-rw-r--r-- 1 root root 1286 Mar 18 2024 000-default.conf
-rw-r--r-- 1 root root 4558 Aug 24 16:43 default-ssl.conf
user@DESKTOP-PB8DGD8: /etc/ssl/mycert$ ls -l /etc/apache2/sites-enabled/
ls: cannot access '/etc/apache2/sites-enabled/': No such file or directory
user@DESKTOP-PB8DGD8: /etc/ssl/mycert$ ls -l /etc/apache2/sites-enabled/
total 0
-rwxr-xr-x 1 root root 35 Aug 24 14:56 000-default.conf -> ../sites-available/000-default.conf
-rwxr-xr-x 1 root root 35 Aug 24 16:20 default-ssl.conf -> ../sites-available/default-ssl.conf
user@DESKTOP-PB8DGD8: /etc/ssl/mycert$ sudo nano /etc/apache2/sites-available/default-ssl.conf
user@DESKTOP-PB8DGD8: /etc/ssl/mycert$ sudo apache2ctl configtest
syntax OK
user@DESKTOP-PB8DGD8: /etc/ssl/mycert$ sudo systemctl restart apache2
system has not been booted with systemd as init system (PID 1). Can't operate.
failed to connect to bus: Host is down
user@DESKTOP-PB8DGD8: /etc/ssl/mycert$ sudo service apache2 restart
user@DESKTOP-PB8DGD8: /etc/ssl/mycert$ apache2
user@DESKTOP-PB8DGD8: /etc/ssl/mycert$ sudo service apache2 status
* apache2 is running
user@DESKTOP-PB8DGD8: /etc/ssl/mycert$
```

Step 11: Test Secure HTTPS Communication

Open a browser and visit:

<https://localhost>

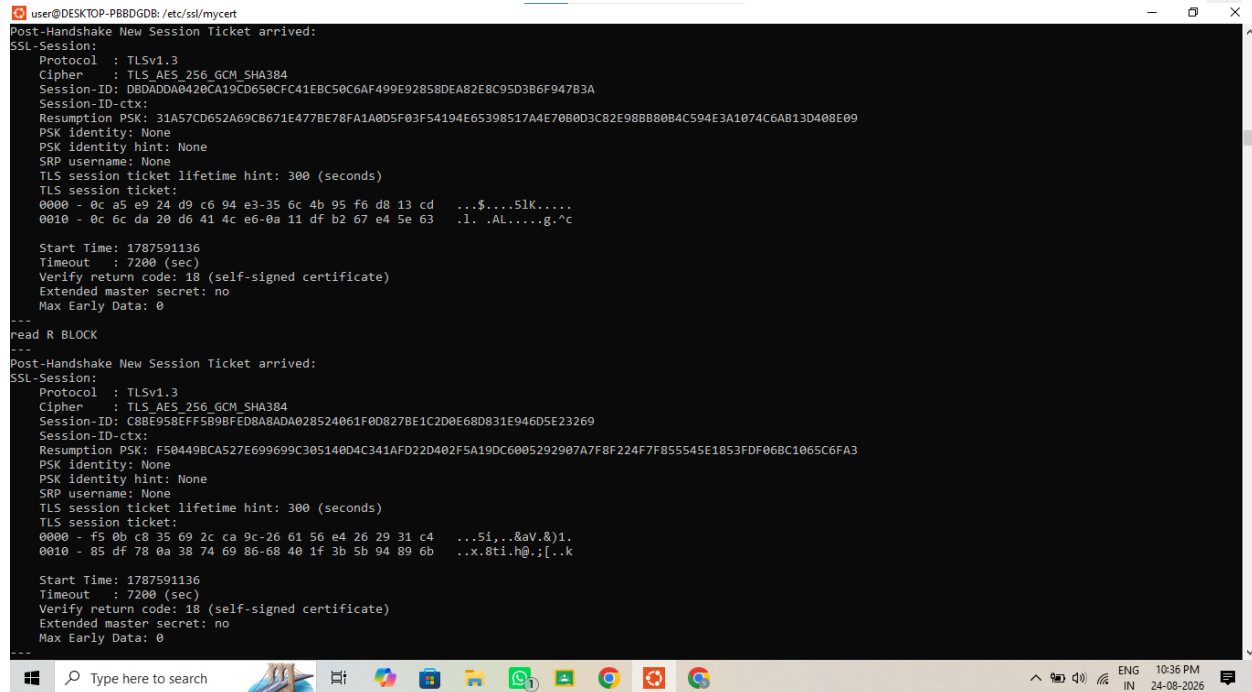


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Step 12: Verify the TLS Connection

Use OpenSSL to test the secure connection:

`openssl s_client -connect localhost:443`



```
user@DESKTOP-P8BDGDB: /etc/ssl/mycert
Post-Handshake New Session Ticket arrived:
SSL-Session:
    Protocol : TLSv1.3
    Cipher   : TLS_AES_256_GCM_SHA384
    Session-ID: DBDADD0A0420CA19CD650CF41EBC50C6AF499E92858DEA82E8C95D3B6F94783A
    Session-ID-ctx:
    Resumption PSK: 31A57CD652A69C8671E477BE78FA1A0D5F03F54194E65398517A4E70B0D3C82E98B80B4C594E3A1074C6AB13D408E09
    PSK identity hint: None
    SRP username: None
    TLS session ticket lifetime hint: 300 (seconds)
    TLS session ticket:
    0000 - 0c a5 e9 24 d9 c6 94 e3-35 6c 4b 95 f6 d8 13 cd ...$.5LK.....
    0010 - 0c 6c da 20 d6 41 4c e6-0a 11 df b2 67 e4 5e 63 ..l..AL.....g.^c

    Start Time: 1787591136
    Timeout    : 7200 (sec)
    Verify return code: 13 (self-signed certificate)
    Extended master secret: no
    Max Early Data: 0
---
read R BLOCK
---
Post-Handshake New Session Ticket arrived:
SSL-Session:
    Protocol : TLSv1.3
    Cipher   : TLS_AES_256_GCM_SHA384
    Session-ID: C8BE958EFF5B9BFED8A8ADA028524061F00827BE1C2D0E68D831E94605E23269
    Session-ID-ctx:
    Resumption PSK: F504498CA527E699699C30514804C341AFD22D402F5A19DC6005292907A7F8F224F7F855545E1853FDF068C1065C6FA3
    PSK identity hint: None
    SRP username: None
    TLS session ticket lifetime hint: 300 (seconds)
    TLS session ticket:
    0000 - f5 0b c8 35 69 2c ca 9c-26 61 56 e4 26 29 31 c4 ...5i,..&av.&)1.
    0010 - 85 df 78 0a 38 74 69 86-68 40 1f 3b 5b 94 89 6b ..x.8ti.h@.;[.k

    Start Time: 1787591136
    Timeout    : 7200 (sec)
    Verify return code: 18 (self-signed certificate)
    Extended master secret: no
    Max Early Data: 0
---
```

Step 13: Check the Certificate

Display certificate information:

`openssl x509 -in /etc/ssl/mycert/server.crt -text -noout`

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```
user@DESKTOP-PB8DGOB: /etc/ssl/mycert
read R BLOCK
closed
user@DESKTOP-PB8DGOB:/etc/ssl/mycert$ openssl x509 -in /etc/ssl/mycert/server.crt -text -noout
Certificate:
    Data:
        Version: 3 (0x2)
        Serial Number:
            50:08:fd:92:15:e7:a8:7b:4b:ba:6f:5f:86:30:8f:57:07:c3:1a:1a
        Signature Algorithm: sha256WithRSAEncryption
        Issuer: C = IN, ST = Maharashtra, L = Mumbai, O = My college, OU = CS department, CN = localhost, emailAddress = yzx@gmail.com
        Validity
            Not Before: Aug 24 16:20:03 2026 GMT
            Not After : Aug 24 16:20:03 2027 GMT
        Subject: C = IN, ST = Maharashtra, L = Mumbai, O = My college, OU = CS department, CN = localhost, emailAddress = yzx@gmail.com
        Subject Public Key Info:
            Public Key Algorithm: rsaEncryption
            Public-Key: (2048 bit)
            Modulus:
                00:9d:22:99:61:e4:de:98:27:61:2c:3c:1c:55:7c:
                15:0a:6e:d6:80:52:8b:8d:56:19:47:56:3f:a4:c6:
                a6:4d:43:ba:37:41:68:bd:8e:3b:57:6d:cf:4e:0c:
                4f:63:09:4d:9a:a4:00:8f:e8:a5:ac:f0:4d:41:a4:
                91:8b:ed:cf:65:b8:78:5e:d5:be:fd:fb:b4:56:20:
                91:c8:40:6d:ee:50:19:45:b0:45:6b:6d:04:75:b2:
                96:74:44:6c:26:9b:ae:dd:19:0c:c2:0a:22:91:3a:
                f4:16:93:17:6a:84:87:27:a6:e1:ae:da:5b:16:39:
                85:a9:bc:ea:4a:3c:0c:b7:c7:b0:14:e0:ac:fc:50:
                c9:86:6c:9e:16:44:c2:dc:4e:f4:f8:f8:cb:72:39:
                b7:fc:76:3f:32:60:a9:b8:59:a3:a7:09:f3:ed:58:
                63:f2:bb:e4:c6:25:30:d4:03:b5:38:42:fb:70:bc:
                24:3a:fd:e1:09:86:04:dc:2a:0a:7f:ee:b3:63:ce:
                b5:2f:05:c1:85:02:63:f1:9f:fa:7a:2f:45:cf:97:
                44:1b:69:ce:8b:a6:38:e5:b5:f3:b9:cd:76:1f:22:
                09:e2:a3:8a:76:19:ce:19:4f:25:af:70:87:59:d5:
                b1:e3:ac:03:90:10:5d:80:98:b6:4b:e4:5b:25:a7:
                f2:89
            Exponent: 65537 (0x10001)
        X509v3 extensions:
            X509v3 Subject Key Identifier:
                A0:0C:52:BA:E7:AC:53:C3:CC:F1:98:38:21:7D:45:C3:84:19:29:4C
            X509v3 Authority Key Identifier:
                A0:0C:52:BA:E7:AC:53:C3:CC:F1:98:38:21:7D:45:C3:84:19:29:4C
            X509v3 Basic Constraints: critical
                CA:TRUE
    Signature Algorithm: sha256WithRSAEncryption
    Signature Value:
        95:8c:f2:36:62:df:d8:51:f1:39:e7:d0:e0:17:5c:61:76:bf:
        22:41:9b:24:23:d7:32:62:9a:93:e3:24:b4:f3:95:9c:e5:53:
        1e:6c:1d:aa:1d:0c:8b:3f:24:1f:97:56:82:ce:60:a0:f6:94:
        f9:09:42:76:1b:49:f5:a2:eb:bc:12:88:90:7e:54:44:26:90:
        1b:78:81:f9:a7:f6:5d:5d:cf:54:63:09:80:02:f2:e0:f3:1a:
        57:ed:4c:6d:e2:9f:8e:b8:80:f8:09:16:36:ee:e8:20:fd:aa:
        9e:31:4e:8c:89:f2:27:68:4a:d5:4a:77:2b:c0:37:7a:b4:cb:
        38:e6:03:21:62:ad:7e:61:92:aa:ef:e8:81:81:1d:04:63:d5:
        3a:85:c4:06:6e:96:2f:4f:17:31:50:3c:4a:10:f0:28:4b:af:
        dd:91:02:08:ff:cc:38:ec:95:0e:b8:80:b1:bd:2e:cb:98:59:
        cd:b9:94:33:3b:a2:28:42:bb:b0:bf:d8:89:91:26:35:7f:6e:
        db:b9:13:6c:9d:96:df:94:cd:27:6f:bc:ad:0c:d6:bc:d7:23:
        8e:b9:d0:54:a3:d2:9c:bc:f0:a2:51:b5:6e:80:c0:f0:f0:dc:
        1c:3a:3b:7a:bc:60:45:ce:77:c0:8f:d6:dd:02:5e:55:87:60:
        17:14:fe:a5
user@DESKTOP-PB8DGOB:/etc/ssl/mycert$
```

Step 14: Secure Certificate Management

Check certificate expiry date:

```
openssl x509 -enddate -noout -in /etc/ssl/mycert/server.crt
```

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```
user@DESKTOP-PB8DGOB: /etc/ssl/mycert$ openssl x509 -enddate -noout -in /etc/ssl/mycert/server.crt
c0:86:6e:19e:16:44:c2:dc:4e:f4:f8:f8:cb:72:39:
b7:fc:76:3f:32:60:a9:b8:59:a3:a7:09:f3:ed:58:
63:f2:bb:e4:c6:25:30:d4:03:b5:38:42:fb:70:bc:
24:3a:fd:e1:09:86:04:dc:2a:0a:7f:ee:b3:63:ce:
b5:2f:05:c1:85:02:63:f1:9f:fa:7a:2f:45:cf:97:
44:1b:69:ce:8b:a6:38:e5:b5:f3:b9:cd:76:1f:22:
09:e2:a3:8a:76:19:ce:19:4f:25:af:70:87:59:d5:
b1:e3:ac:03:90:10:5d:80:98:b6:4b:e4:5b:25:a7:
f2:89
Exponent: 65537 (0x10001)
X509v3 extensions:
  X509v3 Subject Key Identifier:
    A0:0C:52:BA:E7:AC:53:C3:CC:F1:98:38:21:7D:45:C3:84:19:29:4C
  X509v3 Authority Key Identifier:
    A0:0C:52:BA:E7:AC:53:C3:CC:F1:98:38:21:7D:45:C3:84:19:29:4C
  X509v3 Basic Constraints: critical
    CA:TRUE
Signature Algorithm: sha256WithRSAEncryption
Signature Value:
  95:8c:f2:36:62:df:d8:51:f1:39:e7:d0:e0:17:5c:61:76:bf:
  22:41:9b:24:23:d7:32:62:9a:93:e3:24:b4:f3:95:9c:e5:53:
  1e:0c:1d:aa:1d:0c:8b:3f:24:1f:97:56:82:ce:d0:a0:f0:94:
  f9:00:42:76:1b:a9:f5:a2:eb:bc:12:88:90:76:54:44:26:99:
  1b:70:81:f9:a7:f6:5d:5d:ef:54:63:9b:0b:02:f2:e0:f3:1a:
  57:ed:4c:6d:e2:9f:8e:b8:80:f8:09:16:36:ee:e8:20:fd:aa:
  0e:31:4e:8c:80:f2:27:68:4a:d5:4a:77:2b:c0:37:7a:b4:cb:
  38:e6:03:21:62:ad:7e:61:92:aa:ef:e8:81:81:1d:04:63:d5:
  3a:85:c4:06:6e:96:2f:4f:17:31:50:3c:4a:10:f0:28:4b:af:
  dd:91:02:08:ff:cc:38:ec:95:0e:b8:80:b1:bd:2e:cb:98:59:
  cd:b9:94:33:b3:a2:28:42:bb:b0:bf:d8:89:91:26:35:7f:6e:
  db:38:13:6c:9d:96:df:94:cd:27:6f:bc:ad:0c:d6:bc:d7:23:
  8e:b9:d0:54:a3:d2:9c:bc:f0:a2:51:b5:6e:80:c0:f0:d0:dc:
  1c:3e:3b:7a:bc:60:45:ce:77:c0:8f:d6:dd:02:5e:55:87:60:
  17:14:fe:a5
user@DESKTOP-PB8DGOB: /etc/ssl/mycert$
notAfter=Aug 24 16:20:03 2027 GMT
user@DESKTOP-PB8DGOB: /etc/ssl/mycert$
```

Step 16: Configure Strong TLS Security

Edit the SSL configuration:

`sudo nano /etc/apache2/mods-available/ssl.conf`

```
user@DESKTOP-PB8DGOB: /etc/ssl/mycert$ nano /etc/apache2/mods-available/ssl.conf
GNU nano 7.2 /etc/apache2/mods-available/ssl.conf
# Clients may prefer lower grade encryption.  You should enable this
# option if you want to enforce stronger encryption, and can afford
# the CPU cost, and did not override SSLCipherSuite in a way that puts
# insecure ciphers first.
# Default: Off
#SSLHonorCipherOrder on

# The protocols to enable.
# Available values: all, SSLv3, TLSv1, TLSv1.1, TLSv1.2
# SSL v2 is no longer supported
SSLProtocol -all +TLSv1.2 +TLSv1.3

# Allow insecure renegotiation with clients which do not yet support the
# secure renegotiation protocol. Default: Off
#SSLInsecureRenegotiation on

# Whether to forbid non-SNI clients to access name based virtual hosts.
# Default: Off
#SSLStrictSNIVHostCheck On

# Warning: Session Tickets require regular reloading of the server!
# Make sure you do this (e.g. via logrotate) before changing this setting!
SSLSessionTickets off
```

Restart Apache:

SHETH L.U.J. AND SIR M.V. COLLEGE

sudo systemctl restart apache2

```
user@DESKTOP-PBBDGDB: /etc/ssl/mycert
b7:fc:76:3f:32:60:a9:b8:59:a3:a7:09:f3:ed:58:
63:f2:bb:e4:c6:25:30:d4:03:b5:38:42:fb:70:bc:
24:3a:fd:e1:80:80:04:dc:2a:0a:7f:ee:b3:63:ce:
b5:2f:05:cd:85:02:b3:ff:9f:fa:7a:2f:d5:cf:97:
44:1b:69:ce:8b:a6:38:e5:b5:f3:b0:cd:76:1f:22:
09:e2:a3:8a:76:19:ce:19:4f:25:af:70:87:59:d5:
b1:e3:ac:03:90:10:5d:80:98:b6:4b:e4:5b:25:a7:
f2:89
Exponent: 65537 (0x10001)
X509v3 extensions:
X509v3 Subject Key Identifier:
A0:0C:52:BA:E7:AC:53:C3:CC:F1:98:38:21:7D:45:C3:84:19:29:4C
X509v3 Authority Key Identifier:
A0:0C:52:BA:E7:AC:53:C3:CC:F1:98:38:21:7D:45:C3:84:19:29:4C
X509v3 Basic Constraints: critical
CA:TRUE
Signature Algorithm: sha256WithRSAEncryption
Signature Value:
95:8c:f2:36:62:df:d8:51:f1:39:e7:d0:e0:17:5c:61:76:bf:
22:41:9b:24:23:d7:32:62:9a:93:e3:24:b4:f3:95:9c:e5:53:
1e:6c:1d:aa:1d:0c:8b:3f:24:1f:97:56:82:ce:60:a0:f6:94:
f9:00:42:76:1b:49:f5:a2:eb:bc:12:88:90:7e:54:44:26:90:
1b:78:81:f9:a7:f6:5d:5d:cf:54:63:9b:8b:82:f2:e0:f3:1a:
57:ed:4c:6d:e2:9f:8e:b8:80:f8:09:16:36:ee:e8:20:fd:aa:
9e:31:4e:8c:89:f2:27:68:4a:d5:4a:77:2b:c0:37:7a:b4:cb:
38:e0:03:21:62:ad:7e:61:92:aa:ef:e8:81:81:1d:04:63:d5:
3a:85:c4:06:0e:96:2f:4f:17:31:50:3c:4a:10:f0:28:db:af:
dd:91:02:08:ff:cc:38:ec:05:0e:b0:80:bb:1d:2e:cb:08:59:
cd:b9:94:33:b3:a2:28:42:bb:b0:bf:d0:09:12:26:35:7f:6e:
db:38:13:6c:9d:96:df:94:cd:27:6f:bc:ad:0c:d6:bc:d7:23:
8e:b9:d0:54:a3:d2:9c:bc:f0:a2:51:b5:6e:80:c0:f0:f0:dc:
1c:3e:3b:7a:bc:60:45:ce:77:c0:8f:d6:dd:02:5e:55:87:60:
17:14:fe:a5
user@DESKTOP-PBBDGDB:/etc/ssl/mycert$ openssl x509 -enddate -noout -in /etc/ssl/mycert/server.crt
notAfter=Aug 24 16:20:03 2027 GMT
user@DESKTOP-PBBDGDB:/etc/ssl/mycert$ sudo nano /etc/apache2/mods-available/ssl.conf
user@DESKTOP-PBBDGDB:/etc/ssl/mycert$ sudo systemctl restart apache2
System has not been booted with systemd as init system (PID 1). Can't operate.
Failed to connect to bus: Host is down
user@DESKTOP-PBBDGDB:/etc/ssl/mycert$ sudo service apache2 restart
* Restarting Apache httpd web server apache2
[ OK ]
user@DESKTOP-PBBDGDB:/etc/ssl/mycert$ sudo service apache2 status
* apache2 is running
user@DESKTOP-PBBDGDB:/etc/ssl/mycert$
```

7th cyber ubuntu - Google Doc x Apache2 Ubuntu Default Page: x + Ask Gemini | - x

Not secure https://localhost



Apache2 Default Page

Ubuntu

It works!

This is the default welcome page used to test the correct operation of the Apache2 server after installation on Ubuntu systems. It is based on the equivalent page on Debian, from which the Ubuntu Apache packaging is derived. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should **replace this file** (located at `/var/www/html/index.html`) before continuing to operate your HTTP server.

If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.

Configuration Overview

Ubuntu's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Ubuntu tools. The configuration system is **fully documented in `/usr/share/doc/apache2/README.Debian.gz`**. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the **manual** if the `apache2-doc` package was installed on this server.

The configuration layout for an Apache2 web server installation on Ubuntu systems is as follows:

```
/etc/apache2/
|-- apache2.conf
|   |-- ports.conf
|-- mods-enabled
|   |-- *.load
|   |-- *.conf
|-- conf-enabled
```

Type here to search

ENG 10:47 PM 24-08-2026